RV College of Engineering®

SKANDA P R <skandapr.cs22@rvce.edu.in>

Re: DPR Project

1 message

Jyothi Shetty <jyothis@rvce.edu.in> Fri, Apr 10, 2026 at 5:16 AM
To: Roger Dev <Roger.Dev@roots-of-unity.org>, SKANDA P R <skandapr.cs22@rvce.edu.in>

Sure Roger, thank you.

@SKANDA P R please follow the trailing mail.

Thanks & Regards

Dr. Jyoti Shetty
Associate Professor
Department of Computer Science and Engineering
RV College of Engineering
https://rvce.edu.in/department/cse/dr_jyoti_shetty/

On Fri, Apr 10, 2026, 4:37 AM Roger Dev <Roger.Dev@roots-of-unity.org> wrote:

> Dear Professor Shetty,
>
> The students on the DPR project asked about how to do qualitative evaluation of the project results. That is not an easy thing to do. I have put together some ideas on the topic below, but then it occurred to me that that is an excellent question to explore using DPR. I recommend that they use the following as a starting point, but refine it via an extended DPR session. I will be curious to see what they come up with! Kindly forward this to the team.
>
> Roger
>
> ---
>
> # DPR Evaluation: Framing and Quantification Approach
>
> ## 1. Framing
>
> We evaluate DPR as a **protocol for reducing hallucinated content through multi-agent cross-checking**, comparing against standard **1:1 human–AI interaction**.
>
> The central hypothesis is:
>
>> **Structured, multi-agent iteration improves reliability by surfacing, challenging, and correcting unsupported claims more effectively than a single interaction pathway.**
>
> DPR is therefore evaluated not just on final outputs, but on how reasoning evolves across rounds.
>
> ---

2. Experimental Setup

For each task:

- **Condition A (Baseline):** 1:1 Human–AI interaction
- **Condition B (DPR):** Human (conductor) + multiple AI agents

Control for:

- identical prompts
- comparable iteration depth (fixed rounds or convergence)
- similar human involvement

Each condition produces:

- initial response
- iterative refinements
- final output

---

3. Core Metrics

1. Hallucination Rate (HR)

> Percentage of claims that are incorrect or unsupported

- Measured on final outputs
- Optionally tracked across intermediate rounds

Primary hypothesis test:

$$
HR_{DPR} < HR_{1:1}
$$

---

2. Hallucination Impact (HI)

> Severity of hallucinated claims in final output

Categorize hallucinations as:

- Minor (no effect on conclusion)
- Moderate (weakens reasoning)
- Major (alters or invalidates conclusion)

Goal: DPR reduces **high-impact errors**, not just total count

---

3. Correction Yield (CY)

> Fraction of hallucinations introduced during iteration that are corrected before final output

$$
CY = \frac{\text{Errors corrected}}{\text{Errors introduced}}
$$

Interpretation:
Measures DPR's ability to *detect and resolve* errors, not just avoid them

---

4. Refinement Gain (RG)

> Improvement from initial answer → final answer

Measured via:

- structured rubric (correctness, completeness, clarity), or
- blind pairwise comparison

$$
RG = \text{Final Quality} - \text{Initial Quality}
$$

Key idea: DPR should improve answers through iteration

---

5. Exploratory Diversity (ED)

> Number of distinct, relevant solution approaches considered before convergence

Count only approaches that are:

- non-redundant
- substantively different
- relevant to the task

Interpretation:
Measures whether DPR expands the solution space before refining

---

4. Evaluation Methodology

Blind Assessment

Evaluators should not know which condition produced each output.

Pairwise Comparison Preferred

Where possible, use:

> "Which answer is better?"

instead of absolute scoring.

Process Tracking

Record:

- intermediate rounds
- emergence and correction of errors
- introduction of alternative approaches

This enables analysis of DPR as a **reasoning process**, not just an output generator.

---

5. Interpretation of Results

DPR is supported if it demonstrates:

- Lower hallucination rate
- Reduced severity of hallucinations
- Higher correction yield
- Positive refinement gain
- Meaningful exploration prior to convergence

Even if:

- convergence takes more rounds
- or final performance is comparable rather than uniformly superior

---

6. Summary

DPR is evaluated as a **protocol for improving reliability through iterative multi-agent reasoning**.

The focus is not solely on final answers, but on whether structured interaction:

- reduces unsupported claims,
- improves reasoning through correction, and
- explores solution space more effectively before converging.